

Assembly ambiguity is not determined by branch statistics: critical groups of de Bruijn graphs

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Abstract

The number of arc-rooted circular sequences sharing a given k -mer spectrum is, by the BEST theorem, the product of a *local* factor $\prod_v (d^+(v) - 1)!$, which depends only on the out-degree multiset of the de Bruijn graph, and a *global* factor, which is the order of the graph's critical (sandpile) group. We make four points about the global factor. First, it is not determined by the degree data: we exhibit two sequences of length 60 whose de Bruijn graphs at $k = 4$ have identical vertex counts, identical out-degree multisets and hence identical local factors, but admit 4 and 256 reconstructions — a factor of 64. This is a strict sharpening of the observation of Kingsford, Schatz and Pop that repeat counting is insufficient. Second, the group structure reads off repeat parameters: a word of length L occurring r times in an otherwise k -unique sequence contributes $\mathcal{K}(G_k) \cong (\mathbb{Z}/r)^{L-k}$. Third — and this is new in this version — the group detects how repeats *interleave*: for two repeats of length L occurring twice each, the disjoint and nested arrangements $A A B B$ and $A B B A$ have isomorphic compacted graphs and $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-2}$, the direct sum of the two single-repeat contributions, while the interleaved arrangement $A B A B$ has $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-1}$, one extra $\mathbb{Z}/2$ that survives to $k = L$ after both individual contributions have died. The degree data are identical in all three cases. Fourth, the invariant is cheap where it matters: contracting unitigs leaves the critical group unchanged, so it is computed on the compacted graph. We demonstrate this on a real genome: for bacteriophage ϕ X174 (5386 bp) the $k = 10$ graph has 5339 vertices but 47 branch points, and its critical group, of order 4.4×10^{12} , is computed in ten milliseconds; the spectrum collapses through 132, 2, 1 at $k = 11, 12, 13$, one past the longest exact repeat. All claims are verified by machine: 30 checks, exact integer arithmetic except where labelled.

1 Introduction

Given a collection of k -mers with multiplicities — an idealised sequencing experiment — how many genomes are consistent with it? The k -mers and their overlaps form a de Bruijn multigraph G_k in which the genome is an Eulerian circuit, so the answer is the number of Eulerian circuits, which the BEST theorem [1, 14] evaluates as

$$\text{ec}(G_k) = |\mathcal{K}(G_k)| \cdot \prod_v (d^+(v) - 1)!, \quad (1)$$

$\mathcal{K}(G_k)$ being the group of spanning arborescences — the *critical* or *sandpile* group. Kingsford, Schatz and Pop [7] computed this for real prokaryotic genomes as a measure of assembly complexity; Loukides, Pissis and co-authors [10] gave fast algorithms for assessing it.

Equation (1) splits the ambiguity into two factors of different character. The product of factorials is a function of the out-degree multiset alone — the *branch statistics*, read off the graph by inspection. The other factor is not. This paper is about what the other factor knows.

What is already known. Two literatures meet here, and we claim nothing in either. On the combinatorial side the critical groups of *complete* de Bruijn graphs are completely understood: Knuth [8] gave the line-digraph recursion, Levine [9] the group structure $\mathcal{K}(\text{DB}(2, n)) \cong \bigoplus_{j=1}^{n-1} (\mathbb{Z}/2^j)^{2^{n-1-j}}$, and Chan, Hollmann and Pasechnik [4] the generalised de Bruijn and Kautz cases. On the assembly side, counting reconstructions via BEST is established [7, 10], as are uniqueness and safety criteria [12] and feasibility thresholds [2, 11]. What appears not to have happened is that the two ever met: we have found no work applying the critical group of a graph to assembly, and none that looks at the *structure* of $\mathcal{K}(G_k)$ for a genome de Bruijn graph, which is a subgraph of a complete one and not covered by [9, 4].

What is new here.

- (i) **Theorem 10:** the global factor is not a function of the degree data. Two explicit sequences have identical branch statistics and reconstruction counts differing by $64\times$.
- (ii) **Theorem 5:** for a single repeat of length L and multiplicity r , $\mathcal{K}(G_k) \cong (\mathbb{Z}/r)^{L-k}$.
- (iii) **Theorem 6** (new in this version): for two repeats the group is additive in the disjoint and nested arrangements and picks up exactly one extra $\mathbb{Z}/2$ in the interleaved arrangement. Since interleaved repeats are the notoriously hard case for assembly, this is the first place the algebra and the practice visibly agree. Proposition 8 then shows that for three repeats the correction is the sandpile group of a small digraph and not a chord-diagram rank — it can be $\mathbb{Z}/3$.
- (iv) **Lemma 4** makes the invariant computable on the compacted graph, and Section 8 computes it on a real genome.

Relation to [7]. Kingsford, Schatz and Pop already argue that repeat counting is insufficient, with the example that three repeats in two copies each (8 reconstructions) is easier than one repeat in six copies (720). Their two situations have *different* out-degree multisets, $(2, 2, 2)$ against (6) , so the local factor already distinguishes them. Theorem 10 fixes the out-degree multiset and still finds a 64-fold spread, and Theorem 6 says what the residual information is: not how many repeats or how many copies, but how they are wired to one another.

2 Setting

Let S be a circular sequence of length N over an alphabet of size σ . For $k \geq 1$ the *de Bruijn multigraph* $G_k = G_k(S)$ has as vertices the distinct k -mers of S and one arc for each of the N occurrences of a $(k+1)$ -mer, from its length- k prefix to its length- k suffix. S traverses every arc once, so G_k is connected and balanced, and S is an Eulerian circuit; every Eulerian circuit is a

circular sequence with the same k -mer spectrum, counted with multiplicities. We write $n_k = |V(G_k)|$, $d^+(v)$ for out-degree (the number of occurrences of v), and $\text{Loc}(G) := \prod_v (d^+(v) - 1)!$.

Remark 1 (Counting convention). Throughout, reconstruction counts are *arc-rooted*: circuits are counted as beginning with a fixed arc, which is the convention under which (1) holds [1, 14]. Circular sequences up to cyclic shift are counted by dividing by N when S is aperiodic, and by Burnside's lemma otherwise; none of the algebra below depends on which convention is chosen, since the correction is a function of N alone.

The *critical group* is $\mathcal{K}(G) := \text{coker}(\Delta^\top)$, Δ the out-Laplacian $D^+ - A$ with the row and column of a fixed root deleted; for a connected balanced digraph the isomorphism class does not depend on the root, and $|\mathcal{K}(G)|$ is the number of spanning arborescences. We compute it as a Smith normal form.

Remark 2 (The tower). $G_{k+1}(S)$ is a spanning subgraph of the line digraph $L(G_k(S))$, with equality for the complete de Bruijn graph; the prefix and suffix maps are the tail and head maps of G_k and are the degeneracy maps of the non-normal tower of [5, V]. The line-digraph recursion of [5, III] specialises to Knuth's formula and reproduces $|\mathcal{K}(\text{DB}(\sigma, k))| = \sigma^{\sigma^k - k - 1}$ (check (T)); check (L) reproduces Levine's decomposition. These are consistency checks on our conventions, not results.

3 The local and global factors

Proposition 3 (Decomposition). *The number of arc-rooted circular sequences with the same k -mer spectrum as S equals $|\mathcal{K}(G_k)| \cdot \text{Loc}(G_k)$. The factor $\text{Loc}(G_k)$ depends only on the multiset $\{d^+(v)\}$; the factor $|\mathcal{K}(G_k)|$ does not.*

The first sentence is (1). The second is Theorem 10.

4 Contraction, and a single repeat

Lemma 4 (Unitig contraction). *Let v be a vertex of a connected balanced multidigraph G with $d^-(v) = d^+(v) = 1$, whose incoming and outgoing arcs are distinct, say $u \rightarrow v$ and $v \rightarrow w$. Let G' be obtained by deleting v and replacing those two arcs by a single arc $u \rightarrow w$. Then $\mathcal{K}(G') \cong \mathcal{K}(G)$. Self-loops may be deleted freely: a loop at v adds 1 to both D_{vv}^+ and A_{vv} and so does not appear in Δ .*

Proof. Root away from v and u . The row of v in Δ is $e_v - e_w$ and the column of v has a single off-diagonal entry, in the row of u ; adding the row of v to the row of u clears it and turns the row of u into its row in $\Delta(G')$, after which the row and column of v form a unit block. Both operations are unimodular. The statement about loops is immediate from $\Delta = D^+ - A$ and is check (P). $\square$

Iterating the lemma yields the *compacted* (unitig) de Bruijn graph, in which every maximal non-branching path is one arc and only branch vertices survive.

Theorem 5 (A repeat of length L and multiplicity r). *Let W have length L and let S be circular with every k -mer occurring once except the $L - k + 1$ k -mers of W , each occurring exactly $r \geq 2$ times. Then for $1 \leq k \leq L$*

$$\mathcal{K}(G_k(S)) \cong (\mathbb{Z}/r)^{L-k}.$$

Proof. Put $s = L - k + 1$ and let $v_1, \dots, v_s$ be the k -mers of W in order. Every other vertex has in- and out-degree 1; contract them. What remains is the spine $v_i \rightarrow v_{i+1}$ with multiplicity r and r arcs $v_s \rightarrow v_1$. Rooted at v_1 , on $v_2, \dots, v_s$ the out-Laplacian is $r(I - N)$ with N the nilpotent shift; $I - N$ is unimodular, so the cokernel is $(\mathbb{Z}/r)^{s-1}$. $\square$

5 Two repeats: the group sees interleaving

Fix two distinct words A, B of length L and a circular sequence in which each occurs exactly twice, non-overlapping, with every other k -mer unique; put $s = L - k + 1$. The four occurrences can be arranged in three ways around the circle:

disjoint $A A B B$, nested $A B B A$, interleaved $A B A B$.

The k -mer spectra of the three sequences have identical multiplicity profiles, so n_k , the out-degree multiset, and hence $\text{Loc}(G_k)$ agree. This is Theorem 10's setting one level up, and the question is whether $\mathcal{K}$ can tell them apart.

Theorem 6 (Two repeats). *With $a_1, \dots, a_s$ and $b_1, \dots, b_s$ the k -mers of A and B :*

(i) *the disjoint and nested arrangements have isomorphic compacted graphs, and*

$$\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-2} = (\mathbb{Z}/2)^{s-1} \oplus (\mathbb{Z}/2)^{s-1},$$

the direct sum of the two single-repeat groups of Theorem 5;

(ii) *the interleaved arrangement has*

$$\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-1}.$$

In particular the arrangement is detected by $\mathcal{K}$ and not by the degree data, and at $k = L$ ($s = 1$) the interleaved arrangement retains a $\mathbb{Z}/2$ after both single-repeat contributions have become trivial.

Proof. Contract by Lemma 4. The surviving vertices are the $2s$ shared k -mers, each of in- and out-degree 2. The spine arcs $a_i \rightarrow a_{i+1}$ and $b_i \rightarrow b_{i+1}$ have multiplicity 2. The four spacer paths contract to four connector arcs from $\{a_s, b_s\}$ to $\{a_1, b_1\}$, one per transition between consecutive occurrences around the circle. Reading the cyclic order: disjoint gives $a_s \rightarrow a_1$, $a_s \rightarrow b_1$, $b_s \rightarrow b_1$, $b_s \rightarrow a_1$; nested gives $a_s \rightarrow b_1$, $b_s \rightarrow b_1$, $b_s \rightarrow a_1$, $a_s \rightarrow a_1$ — the same multiset, so the two compacted graphs coincide; interleaved gives $a_s \rightarrow b_1$ twice and $b_s \rightarrow a_1$ twice.

(ii) Root at a_1 . On $a_2, \dots, a_s, b_1, \dots, b_s$ in this order the reduced out-Laplacian is $2(I - N)$, N the shift along that path (both arcs of b_s go to the deleted root). $I - N$ is unimodular, so $\mathcal{K} \cong (\mathbb{Z}/2)^{2s-1}$.

(i) Root at a_1 ; write x_v for the generator of the cokernel presentation at v . The rows of a_s and b_s read $2x_{a_s} - x_{b_1}$ and $2x_{b_s} - x_{b_1}$, so $x_{b_1} = 2x_{a_s}$ and x_{b_1} can be eliminated. On the remaining $2s - 2$ generators $a_2, \dots, a_s, b_2, \dots, b_s$ every relation is now even — the spine relations $2x_{a_i} - 2x_{a_{i+1}}$ and $2x_{b_i} - 2x_{b_{i+1}}$, the relation $2x_{b_s} - 2x_{a_s}$, and the former row of b_1 , which becomes $4x_{a_s} - 2x_{b_2}$ — so the relation matrix is $2M'$ with

$$M' = \begin{pmatrix} \text{rows } e_{a_i} - e_{a_{i+1}}, & 2 \leq i \leq s-1 \\ & e_{b_s} - e_{a_s} \\ & 2e_{a_s} - e_{b_2} \\ \text{rows } e_{b_i} - e_{b_{i+1}}, & 2 \leq i \leq s-1 \end{pmatrix}.$$

Expanding $\det M'$ along the columns $a_2, \dots, a_{s-1}$ in turn (each has a single entry, $+1$) leaves the $s \times s$ block on columns $(a_s, b_2, \dots, b_s)$ with rows $2e_{a_s} - e_{b_2}, e_{b_i} - e_{b_{i+1}}$ ($2 \leq i \leq s-1$), $e_{b_s} - e_{a_s}$. Expanding along the first row with the cofactor signs written out, $\det M' = 2(-1)^{1+1}M_{11} + (-1)(-1)^{1+2}M_{12} = 2M_{11} + M_{12}$. The minor M_{11} (the remaining rows on columns $b_2, \dots, b_s$) is upper bidiagonal with unit diagonal, so $M_{11} = 1$; the minor M_{12} (the same rows on columns $a_s, b_3, \dots, b_s$) has a single entry -1 in its first column, in its last row, and expanding along that column leaves a lower bidiagonal block with diagonal -1 , the signs combining to $M_{12} = -1$. Hence $\det M' = 2 - 1 = 1$ in this column order, $|\det M'| = 1$ in any order, M' is unimodular and $\mathcal{K} \cong (\mathbb{Z}/2)^{2s-2}$. $\square$

Remark 7 (Reading). For a single repeat the group is trivial at $k = L$: the repeated L -mer is a branch vertex but its two exits lead to different places and there is only one way to thread it. For two interleaved repeats at $k = L$ there is a genuine choice — leave A for the first B or the second — and $\mathbb{Z}/2$ records it. Nested and disjoint repeats offer no such choice: whichever way one threads A , the two B 's are traversed consecutively. This is the algebraic form of the folklore that interleaved repeats are what makes assembly hard. Additivity of $\mathcal{K}$ over repeats, asked for in version 1, therefore holds exactly when the repeats do not interleave; the general correction term is indexed by the interleaving pattern, and Theorem 6 is its two-repeat case.

Check (I) verifies the theorem three ways: on clean random sequence instances (the prediction is exact in every one), on the explicit compacted graphs (six branch vertices; connector multiplicities $\{1, 1, 1, 1\}$ for disjoint and nested and $\{2, 2\}$ for interleaved), and by symbolic Smith normal form of the matrices in the proof for $s \leq 7$.

Three repeats, and what the correction is not

With three repeats the natural guess — suggested to us by a reader — is that the interleaving correction is a function of the chord diagram of the six occurrences through the rank over $\mathbb{F}_2$ of its intersection matrix. It is not.

Proposition 8 (Three repeats). *Let A, B, C be distinct words of length L , each occurring exactly twice, with every other k -mer unique and $s = L - k + 1$. Let D_w be the digraph on the three symbols with one arc from the symbol of each occurrence to the symbol of the next occurrence around the circle, loops deleted. Then $\mathcal{K}(G_L) \cong \mathcal{K}(D_w)$, and according to which pairs of the three chords cross:*

<i>crossing pairs</i>	$\mathbb{F}_2$ -rank of intersection matrix	$\mathcal{K}(G_k), s = 1$	$\mathcal{K}(G_k), s = 2$
<i>none</i>	0	0	$(\mathbb{Z}/2)^3$
<i>one pair</i>	2	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^4$
<i>a path of two</i>	2	$\mathbb{Z}/3$	$(\mathbb{Z}/2)^3 \oplus \mathbb{Z}/3$
<i>all three pairs</i>	2	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^5$

In every case $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{3(s-1)} \oplus \mathcal{K}(D_w)$ for $s = 1, 2$.

Proof. At $s = 1$ the compacted graph is D_w by construction (Lemma 4 deletes the loops), which gives $\mathcal{K}(G_L) = \mathcal{K}(D_w)$; the four values are the sandpile groups of the four 2-regular digraphs on three vertices that occur. For instance the path pattern $ABACBC$ gives the complete symmetric digraph on three vertices, whose sandpile group is $\mathbb{Z}/3$. The $s = 2$ column and the splitting are verified exactly on all 90 circular words with two of each letter (check (M)); we have not proved the splitting for general s . $\square$

Remark 9. Three of the four patterns have intersection rank 2 and give three different groups, one of them of odd order: multiplicity-two repeats can interleave into 3-torsion. So the correction is not a chord-diagram rank, nor even a 2-group. The right object is the sandpile group of the connector digraph D_w , which for two repeats is trivial or $\mathbb{Z}/2$ (Theorem 6) and for three is the table above.

6 Branch statistics are insufficient

Theorem 10 (The global factor is not degree-determined). *There exist circular sequences $S_{\text{lo}}, S_{\text{hi}}$ of length 60 over $\{A, C, G, T\}$ such that at $k = 4$ their de Bruijn graphs have the same number of vertices $n_4 = 51$ and the same out-degree multiset (seven branch vertices of out-degree 2 and one of out-degree 3), hence the same local factor $\text{Loc} = 2$, but*

$$\mathcal{K}(G_4(S_{\text{lo}})) = \mathbb{Z}/2, \quad \mathcal{K}(G_4(S_{\text{hi}})) = (\mathbb{Z}/2)^4 \oplus \mathbb{Z}/8,$$

so that the numbers of reconstructions are 4 and 256. Explicit witnesses:

$S_{\text{lo}}:$ **GAACGAGAGAAAAACACGTACCAGCGCGGGGGCGGGACGGCAGGTGACGCTCTGGGAAG**
 $S_{\text{hi}}:$ **GCACAGGGTCCAGTCTGCGAACGCGCTACAGTCTGGGTTCAAGCGGTCCTCTGTGCATCG**

Proof. Exact computation; check (N). □

Corollary 11. *No function of the out-degree multiset of G_k determines the number of reconstructions.*

In a randomised search over 1500 sequences of length 60 at $k \in \{4, 5\}$, 67 of the 98 degree-classes containing more than one sequence contained more than one critical group (check (N), sampled).

7 Computation

Computing $\mathcal{K}(G_k)$ directly is one Smith normal form of an $(n_k - 1) \times (n_k - 1)$ integer matrix, $O(n_k^3)$ with the usual bit-length control [5, I], which for a genome is impractical. Lemma 4 removes the difficulty: the critical group of G_k equals that of its compacted graph, whose vertices are the branch points. The compaction is linear in N — one pass to find the branch vertices, one walk per outgoing arc — and assemblers already build this graph. The cost is therefore governed by repeat content, not genome length; the next section shows what this means in practice.

8 A real genome

Bacteriophage φX174 (GenBank NC_001422.1, 5386 bp, circular) was the first genome ever sequenced. Its longest exact circular repeat has length $R = 12$. Table 1 gives the critical-group spectrum around that scale. The compaction is dramatic: at $k = 10$ the graph has 5339 vertices and 47 branch points, and the Smith normal form takes ten milliseconds. The group order at $k = 10$, 4.4×10^{12} , is the global part of the number of 10-mer reconstructions of the phage; it falls to 132 at $k = 11$, to 2 at $k = 12$, and the graph becomes a single cycle at $k = 13 = R + 1$.

Two remarks. The levels $k \leq 9$ have compacted graphs of 146 to 1384 vertices; they are within reach of a sparse or p -adic Smith normal form but not of the dense routine used here, and

k	n_k	branch vertices	$ \mathcal{K}(G_k) $	invariant factors
≤ 9	≤ 5233	≥ 146	—	above the exact-SNF cap used here
10	5339	47	4 408 682 819 584	$2^8, \dots, 4305354316$
11	5376	10	132	2, 66
12	5384	2	2	2
13	5386	0	1	— (single cycle)

Table 1: φ X174, NC_001422.1. Branch vertices are the vertices of the compacted graph. All entries exact (check (F)); the sequence file is distributed with the script and no network access is needed.

we have not pursued them. And $k^* = R + 1$ for this genome. A single isolated repeat collapses at $k = R$ (Theorem 5), so a surviving $\mathbb{Z}/2$ at $k = 12$ is *consistent with* Theorem 6 rather than implied by it; the companion paper [6] settles the question by exhibiting the two reconstructions: the longest 12-mer repeat is interleaved with a second repeated 12-mer, and the residual bit is exactly the one of Theorem 6 at $s = 1$.

9 What is open, and what is false

False. The collapse order is not one more than the longest repeat. With $R(S)$ the longest repeated substring and $k^*(S)$ the least k with $\mathcal{K}(G_k)$ trivial, over 40 random sequences of length 100 we find $k^* = R$ in 26 cases and $k^* = R + 1$ in 13 (check (C)). Exhaustive search over short binary sequences gives minimal witnesses: AACAC has $R = 3$, $k^* = 3$; AACACC has $R = 2$, $k^* = 3$; and homopolymer runs such as AAAA have $k^* < R$, because a repeat that always extends the same way branches nowhere. Theorem 5 explains the first family and Theorem 6 the second, but we have no closed form for k^* in general.

Open. (a) *Additivity beyond two repeats.* Theorem 6 and Proposition 8 identify the correction at $s = 1$ with the sandpile group of the connector digraph D_w of the circular word of occurrences, and show it is not a function of the chord diagram’s intersection rank. We conjecture $\mathcal{K}(G_k) \cong \bigoplus_i (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w)$ for m repeats of multiplicities r_i (with D_w then carrying multiplicities); it is proved for $m \leq 2$ and verified for $m = 3$, $r_i = 2$, $s \leq 2$. (b) *Higher multiplicity.* We have not computed the interleaving correction for repeats of multiplicity $r > 2$. (c) *Double-strandedness.* Real assembly graphs are bidirected. This is a different combinatorial object, and we warn against the tempting decomposition of $\mathcal{K}$ into ± 1 -eigenspaces of the reverse-complement involution: an involution splits an abelian group only after 2 is inverted, and the 2-torsion — which is where all of our examples live, and where inverted repeats would show up — is exactly what such a splitting discards. The tools that survive are the signed and $\mathbb{Z}/2$ -voltage Laplacians [15, 13], with one caveat: reverse complementation reverses arc direction, so the passage to the bidirected graph is not a covering of digraphs and the covering theory of [13] does not apply as it stands. (d) *Errors and coverage.* We assume an exact k -mer spectrum with true multiplicities.

10 Verification

The script `dbg_sandpile.py` runs 30 checks in about two minutes, all passing; its output is archived as `dbg_sandpile_output.txt`. Arithmetic is exact except in the two rows marked

sampled.

key	check	#	mode
(T)	$\text{DB}(\sigma, k+1) = L(\text{DB}(\sigma, k)); \mathcal{K} = \sigma^{\sigma^k - k - 1}$	8/8	exact
(L)	$\mathcal{K}(\text{DB}(2, n))$ reproduces Levine [9], $n \leq 5$	4/4	exact
(E)	BEST (1) against brute-force arc-rooted enumeration	6/6	exact
(R)	Theorem 5, 54 parameter settings	1/1	exact
(I)	Theorem 6: sequences, compacted graphs, symbolic SNF	3/3	exact
(M)	Proposition 8: all 90 circular words of three repeats, $s = 1, 2$	1/1	exact
(N)	Theorem 10 witnesses; degree-class splitting rate	2/2	exact / sampled
(C)	collapse order: statistics; minimal binary witnesses	2/2	sampled / exact
(P)	Lemma 4: contraction and self-loops	2/2	exact
(F)	Table 1: φX174	1/1	exact
total		30/30	

Table 2: The verification battery (`dbg_sandpile.py`).

Provenance and changes in version 2

Unrefereed preprint. The classical inputs are the BEST theorem [1, 14], Knuth’s recursion [8] and the matrix–tree theorem for digraphs; the critical groups of complete de Bruijn graphs [9, 3, 4] are used only as checks. Proposition 3 is a reading of (1); the use of BEST to count genome reconstructions is due to [7]. What we claim as new is Theorems 5, 6 and 10 with Corollary 11, and Lemma 4 with its use in Sections 7–8.

Version 2 adds Theorem 6 (resolving the two-repeat case of the additivity question left open in version 1), Proposition 8 (three repeats, refuting the chord-rank guess), the real-genome computation of Section 8, the counting convention of Remark 1, the self-loop clause of Lemma 4, the minimal collapse witnesses, and the warning about eigenspace decompositions in Open (c). These changes respond to a detailed reading of version 1, for which we are grateful; the referee’s four corrections were all right and are all incorporated.

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